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ORIGINAL PAPER

Automated analysis of perfluorinated compounds in human hair and urine samples by turbulent flow chromatography coupled to tandem mass spectrometry

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Abstract Perfluorinated compounds (PFCs) are ubiquitous contaminants of humans and animals worldwide. PFCs are bioaccumulated because of their affinity for proteins. It has been shown they could have a variety of toxicological effects and cause damage to human health, emphasizing the need for sensitive and robust analytical methods to assess their bioaccumulation in humans. In this paper we report the development and validation of an analytical method for analysis of PFCs in the non-invasive human matrices hair and urine. The method is based on rapid and simple sample pre-treatment followed by online turbulent flow liquid chromatography and tandem mass spectrometry (TFC–LC–MS–MS) for analysis of 21 PFCs. The method was validated for both matrices. Percentage recovery was between 60 and 105 for most compounds in both matrices. Limits of quantification ranged from 0.1 to 9 ng mL^{-1} in urine and from 0.04 to 13.4 in hair. The good performance of the method was proved by investigating the presence of selected PFCs in 24 hair and 30 urine samples from different donors living in Barcelona (NE Spain). The results were indicative of bioaccumulation of these compounds in both types of sample. PFOS and PFOA were most frequently detected in hair and PFBA in urine.

Keywords Perfluorinated compounds · LC–MS–MS · Turbulent flow chromatography · Hair · Urine

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Introduction

Perfluorinated compounds (PFCs) are a wide group of synthetic compounds with high resistance to chemical, thermal, or biological degradation. They have unique properties used for production of oil and water-resistant materials. PFCs are widely used in diverse applications including textiles, packaging for use in the food industry, kitchenware, fire-fighting foams, floor polish, insecticide formulations, cosmetics, adhesives, and components of pharmaceuticals, among others.

Concerns about their persistence and bioaccumulative properties were raised when some of these compounds were found to be ubiquitously distributed in wildlife and human populations worldwide; their long elimination half-lives increased these concerns [1–3]. Production of perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA) has now almost ceased and their use is regulated throughout the world. However, new groups of PFCs are used to replace PFOS and PFOA, and the effects and environmental fate of these new PFCs remain unclear.

PFCs are bioaccumulated because of their affinity for proteins. It has been shown they can disrupt thyroid hormones, proteins, high-density cholesterol, and triglycerides [4–6]. Their involvement in hepatotoxicity, immunotoxicity, and hormone disruption has been demonstrated [5], as also

has their reproductive toxicity and deformity caused in mammals exposed during the early stages of development [7]. In work recently published by the UCLA School of Public Health, it was shown that PFCs could cause infertility in women. The study found that women with high levels of PFCs in the blood conceived later than others whose levels were lower [8]. In addition, it has shown that during pregnancy mother-to-child transfer occurs through the umbilical cord blood or during breast feeding during the first months of life [9].

For these reasons, during recent years several studies have been conducted to assess human exposure to PFCs. Major routes of exposure have been identified as drinking water [10], fish consumption [11, 12], and dust inhalation [13]. Human biomonitoring data are highly important for assessing the human health risk from exposure to PFCs. This exposure has mostly been evaluated by analysis of serum [14] and milk [9, 15], but the human distribution pattern of PFCs remains unclear. In addition, there is a lack of information about possible occurrence in other types of human matrix, for example hair and urine. In addition, because infants are sensitive to exposure to PFCs, development of analytical methods to assess human bioaccumulation in infants using non-invasive samples is urgently required.

Nowadays, liquid chromatography (LC) coupled with unit-mass-resolution tandem mass spectrometry (MS–MS), using triple quadrupoles and ion traps, is the technique of choice for analysis of PFCs. Most of these methods are based on sample pre-treatment followed by off-line purification based on solid-phase extraction (SPE) before chromatographic separation on a C_{18} analytical column; a limited number of studies have been performed using on-line SPE [16].

Turbulent-flow chromatography coupled with tandem mass spectrometry (TFC–LC–MS–MS) has been described as a technique that eliminates time-consuming sample clean-up, and increases productivity with good sensitivity [17]. TFC was introduced in the late 1990s as a technique for direct injection of biological fluids on to a column packed with 50-μm spherical porous particles [18]. Nowadays, TFC can be used as a high-throughput sample-preparation technique that makes use of high flow rates in 0.5 or 1.0 mm internal diameter columns packed with particles in the size range 30–60 μm. Small molecules diffuse more extensively than macromolecules (e.g., proteins, lipids, sugars) and are driven into the pores of the sorbent. Because of the high flow rate, the larger molecules and matrix constituents are flushed to waste and do not have the opportunity to diffuse into the particle pores. The trapped analytes are desorbed from the TFC column and eluted on to the analytical column for further separation and detection.

In the work discussed in this paper we focused on the development and analysis of PFCs in two different non-invasive human matrices: human hair and urine. The main objectives of the work were:

1. to develop and validate a sensitive and robust analytical method based on turbulent flow chromatography and tandem mass spectrometry (TFC–LC–MS–MS) for analysis of 21 PFCs in human hair and 18 PFCs in urine; and
2. to assess the presence of selected PFCs in 24 human hair samples and 30 urine samples.

   As far as we are aware, this is the first method for analysis of PFCs in hair and urine based on TFC–LC–MS–MS. This is also one of the first reports of the PFC content of human urine and of possible bioaccumulation of PFCs in human hair.

Materials and methods

Chemicals

Standard solutions were purchased from Wellington Laboratories, Canada. The standard analytes used in this study were:

1. PFAC-MXB (98% purity in methanol) containing perfluorobutanoic acid (PFBA), perfluoropentanoic acid (PFPeA), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUdA), perfluorododecanoic acid (PFDoA), perfluorotridecanoic acid (PFTrA), perfluorotetradecanoic acid (PFTeA, perfluorohexadecanoic acid (PFHxDA), perfluorooctadecanoic acid (PFODA), perfluorobutanesulfonate (PFBS), perfluorohexanesulfonate (PFHxS), perfluorooctanesulfonate (PFOS), and perfluorodecanesulfonate (PFDS);
2. FTA (98% purity in isopropanol) including perfluorohexyl ethanoic acid (FHEA), perfluorooctyl ethanoic acid FOEA, perfluorodecyl ethanoic acid FDEA; and
3. perfluorooctane sulfonamide (PFOSA) (98% pure in methanol).

Identification and quantification were performed by use of the following internal standards:

1. MPFAC-MXA (>98%) containing [$^{13}C_4$]-perfluorobutanoic acid (MPFBA ($^{13}C_4$)), Ion [$^{18}O_2$]-perfluorohexanesulfonate (MPFHxS ($^{18}O_2$)), [$^{13}C_2$]-perfluorohexanoic acid (MPFHxA ($^{13}C_2$)), Ion [$^{13}C_4$]-perfluorooctanesulfonate (MPFOS ($^{13}C_4$)), , [$^{13}C_4$]-perfluorooctanoic acid

Table 1 Gradient used on the loading pump and the eluting pump for TFC–LC(ESI)–MS–MS analysis. Loading pump A: 0.1% formic acid in water; B: acetone–isopropanol–ACN 10:45:45; C: water; D: Methanol. Eluting pump E: NH_4Ac 20 mmol L^{-1} in water; F: NH_4Ac 20 mmol L^{-1} in methanol

Time (min)	Loading pump					Eluting pump		
	Flow (mL min^{-1})	% solvent				Flow (mL min^{-1})	% phase	
		A	B	C	D		E	F
0:00	1.50	100				0.4	90	10
0:20	0.20			100		0.4	90	10
0:30	0.20	70			30	0.2	90	10
2:30	1.50				100	0.4	20	80
7:30	1.00		100			0.4	10	90
13:00	0.50	20			80	0.4	10	90
14:00	0.50	100				0.4	90	10
16:00	0.50	100				0.4	90	10

(MPFOA ($^{13}C_4$)), [$^{13}C_5$]-perfluorononanoic acid (MPFNA ($^{13}C_5$)), [$^{13}C_2$]-perfluorododecanoic acid (MPFDoA ($^{13}C_2$)), [$^{13}C_2$]-perfluorodecanoic acid (MPFDA ($^{13}C_2$)), [$^{13}C_2$]-perfluoroundecanoic acid (MPFUdA ($^{13}C_2$));

2. MFTA-MXA (>98%) [$^{13}C_2$]-perfluorohexylethanoic acid (MFHEA($^{13}C_2$)), [$^{13}C_2$]-perfluorooctylethanoic acid (MFOEA($^{13}C_2$)), [$^{13}C_2$]-perfluorodecylethanoic acid (MFDEA($^{13}C_2$)); and
3. [$^{13}C_8$]-perfluorooctanesulfonamide (MPFOSA ($^{13}C_8$).

All the mixtures were supplied by Wellington Laboratories, Canada.

Water, methanol, and acetonitrile, Chromasolv Plus for HPLC, ammonium acetate ($AcNH_4$; MW 77.08; 98%), and formic acid (HFo) were obtained from Sigma–Aldrich, Steinheim, Germany.

To eliminate the possibility of cross contamination, polypropylene (PP) insert vials and inert taps were used.

Sampling

Thirty urine and 24 hair samples were collected from donors living in Barcelona (NE Spain), who were individually informed of, and gave their consent to be part of, this study.

Table 2 Precursor ions and transitions (SRM1 used for quantification and SRM2 used for confirmation) with the corresponding collision energies

	Retention time (min)	Precursor ions (*m/z*)	SRM_1 (CE (V))	SRM_2 (CE (V))
PFBA	1.86	213	169 (−10)	119 (−8)
PFPeA	5.47	263	219 (−8)	70 (−8)
PFHxA	8.34	313	269 (−10)	169 (−10)
PFHpA	8.96	363	319 (−10)	169 (−15)
PFOA	9.41	413	169 (−15)	369 (−15)
PFNA	9.73	463	419 (−10)	219 (−15)
PFDA	10.03	513	469 (−14)	169 (−17)
PFUdA	10.41	563	519 (−15)	169 (−15)
PFDoA	10.85	613	569 (−15)	169 (−15)
PFTrDA	11.43	663	619 (−15)	169 (−8)
PFTeDA	12.02	713	669 (−15)	169 (−15)
PFHxDA	13.53	813	769 (−15)	169 (−15)
PFODA	15.04	913	869 (−15)	169 (−15)
PFBS	7.26	299	80 (−80)	99 (−27)
PFHxS	8.99	399	80 (−50)	99 (−50)
PFOS	9.66	499	80 (−49)	99 (−41)
PFDS	10.33	599	80 (−100)	99 (−100)
PFOSA	10.62	498	78 (−40)	64 (−69)
FHEA	9.30	377	293 (−20)	219 (−15)
FOEA	10.13	477	393 (−15)	393 (−15)
FDEA	10.40	577	493 (−15)	493 (−40)

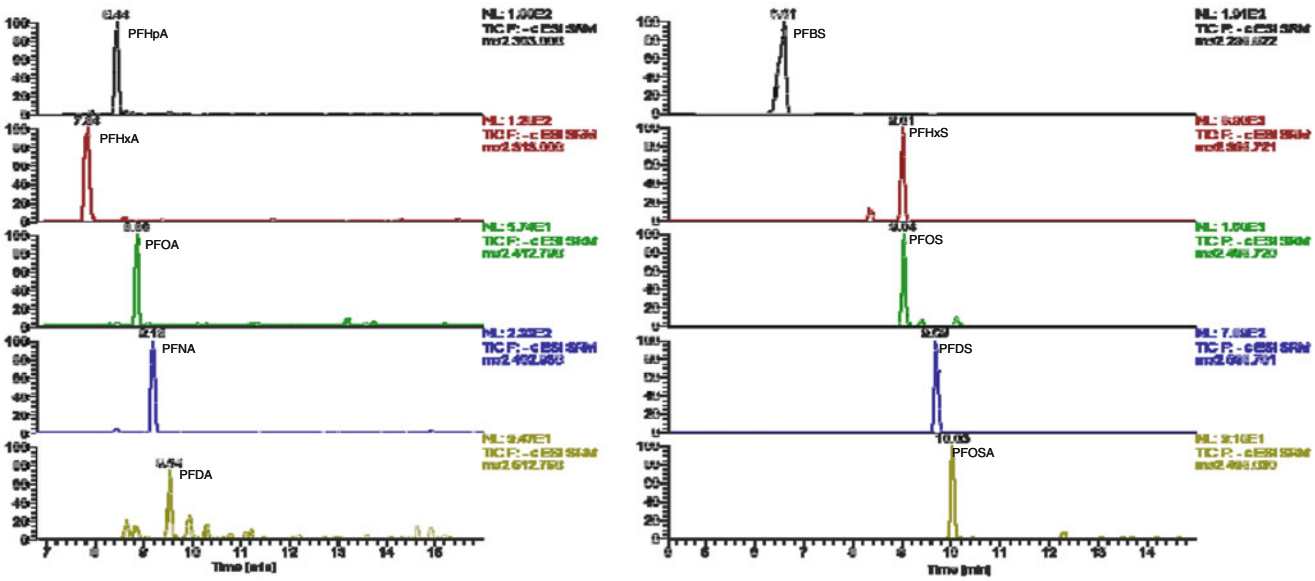

Fig. 1 Chromatogram obtained from analysis of a hair sample fortified with 10 μg kg^{-1} of a standard mixture

Hair samples were cut as close to the scalp as possible. Sample size was sufficient to enable analysis in triplicate. Samples were stored in PP centrifuge tubes and were frozen until the start of the analytical process. Hair samples were collected during December 2010 and January 2011; urine samples were collected in 2009. Urine and hair samples were from different individuals.

Hair and urine samples were previously screened. Samples free from the studied PFCs were selected for use as blank matrices. These samples were used as procedural blanks, to perform spiking experiments, and for construction of matrix-matched calibration curves.

Quantification was performed using the internal standard method, and matrix-matched calibration curves were used to calculate relative response factors.

Equipment

The Aria TLX-1 (Thermo Fisher Scientific, Franklin, MA, USA) turbulent flow chromatograph was used with a PAL autosampler (CTC Analytics, Zwingen, Switzerland), two mixing binary pumps (eluting pump and loading pump), and a three-valve switching device with a six-port valve. The entire system was controlled by Aria software, version 1.6. The Turbo Flow TFC columns used (Thermo Fisher Scientific: Cyclone (50 mm×0.5 mm, 60 μm particle size, 60 Å pore size) and C18 XL (50 mm×0.5 mm, 60 μm particle size, 60 Å pore size)) were connected in tandem . The analytical LC column was a Hypersil Gold PFP 3 mm× 50 mm, 3 μm (Thermo Fisher Scientific). The PFP column was used on the basis of our previous experiments in which the Hypersil Gold PFP column was compared with a normal C_{18} column. The Hypersil Gold PFP column was eventually selected because of its high chromatographic resolution and the separation of matrix interferences. To delay any possible contamination of the system and the solvents, an extra analytical column BDS Hypersil C_8 (50 mm×3 mm; Thermo Scientific) was used after the LC pump.

The triple-quadrupole mass spectrometer was a TSQ Vantage (Thermo Fisher Scientific, San Jose, CA, USA) equipped with an electrospray ionisation source. All analysis was performed in the negative ionisation mode with a spray potential of 2000 V. The sheath, auxiliary, and ion sweep gas used was nitrogen at 30, 15, and 0.5 arbitrary units, respectively. The system was operated in the selected reaction monitoring (SRM) mode with argon as collision gas at a pressure of 1.0 mTorr. Capillary and vaporizer temperatures were 270 °C and 300 °C, and the scan time and width were 0.02 s and 0.02 *m/z*, respectively. Data were processed by use of Xcalibur software version 1.4.

Sample pre-treatment

Hair samples were pre-cleaned to remove possible traces of external contamination. The hair was soaked in water for 15 min. After this time, the water was discarded and the samples were washed twice with acetone, and then air-dried. Hair samples were cut into small pieces and then ground with an agate mortar and pestle till pulverized. Each sample (0.25 g) was then weighed into a PP tube and 0.5 mL acetone was added. This suspension was fortified with 10 μL surrogate internal standard solution in methanol to furnish a final concentration 10 μg L^{-1}. Samples were left to equilibrate for 1 h at room temperature, then 5 mL acetonitrile were added to the PP tubes and the hair was extracted in an ultrasonic bath for 15 min.

Table 3 Validation data: method limits of detection (MLOD), method limit of quantification (MLOQ), decision limit (CCα), detection capability (CCβ), recoveries and repeatability for selected compounds in both type of matrix

	MLOD (ng mL^{-1})	MLOQ (ng mL^{-1})	CCα* (ng mL^{-1})	CCβ* (ng mL^{-1})	Recovery (%)			Inter-day repeatability (RSD, %)	Matrix effect (%) (10 ng mL^{-1})
					5 ng mL^{-1}	10 ng mL^{-1}	50 ng mL^{-1}		
Urine									
PFBA	0.39	1.29	1.36	8.93	50	66	48	10	378
PFPeA	0.74	2.48	2.72	5.31	43	49	53	13	-36
PFHxA	0.01	0.10	0.01	0.78	45	61	58	8	115
PFHpA	0.70	2.35	0.10	3.10	64	54	71	9	345
PFOA	0.02	0.10	0.08	0.23	123	55	69	13	116
PFNA	0.04	0.13	0.12	0.58	119	71	67	19	234
PFDA	0.14	0.45	0.28	0.70	56	61	63	7	101
PFUdA	0.45	1.50	1.76	2.47	73	42	56	5	104
PFDoA	0.29	1.00	0.18	0.99	99	71	83	14	280
PFTrA	0.29	1.00	0.22	1.72	70	59	61	19	140
PFTeA	0.56	1.88	0.39	1.56	66	58	63	17	-85
PFHxDA	0.36	5.00	2.50	5.91	69	43	52	18	-97
PFODA	2.68	8.95	6.96	9.29	–	63	47	18	-99
PFBS	0.74	2.46	0.01	4.17	51	59	46	11	136
PFHxS	0.09	0.31	0.01	2.01	100	70	76	12	231
PFOS	0.11	0.37	0.01	0.17	104	91	94	18	120
PFDS	0.10	0.34	0.04	1.34	104	103	90	13	247
PFOSA	0.07	0.22	0.02	1.23	122	95	95	8	539
Hair									
	MLOD (ng g^{-1})	MLOQ (ng g^{-1})	CCα* (ng g^{-1})	CCβ* (ng g^{-1})	Recovery (%) 5 ng g^{-1}	10 ng g^{-1}	50 ng g^{-1}	100 ng g^{-1}	Inter-day repeatability (%RSD)
PFBA	0.18	0.59	0.87	5.15	133	129	99	84	11
PFPeA	0.28	0.92	0.45	3.19	78	74	103	88	16
PFHxA	0.10	0.34	0.05	1.15	69	69	96	71	17
PFHpA	0.30	0.99	0.03	2.59	60	59	69	54	15
PFOA	0.10	0.34	0.08	0.26	88	83	76	85	16
PFNA	0.01	0.04	1.25	1.71	55	60	58	64	8
PFDA	0.02	0.06	0.82	1.38	73	103	82	81	13
PFUdA	0.50	1.67	1.66	2.38	74	74	72	79	19
PFDoA	1.29	4.28	3.28	6.13	74	83	85	81	19
PFTrDA	1.00	3.34	0.30	1.81	98	96	91	88	14
PFTeDA	2.00	6.65	2.32	5.78	–	90	82	95	18
PFHxDA	2.50	8.34	2.64	6.46	–	70	69	71	17
PFODA	4.00	13.34	8.47	10.36	–	–	60	63	20
PFBS	0.09	0,28	0.01	4.01	52	54	58	57	26
PFHxS	0.08	0,25	0.01	2.36	67	61	64	54	16
PFOS	0.08	0,25	0.01	0.19	61	62	61	54	13
PFDS	0.02	0,06	0.24	1.67	63	30	43	40	11
PFOSA	0.05	0,17	0.16	1.42	67	19	172	96	17
FHEA	0.12	0.40	0.01	2.54	29	112	115	132	22
FOEA	0.14	0.45	0.02	1.45	107	73	50	50	11
FDEA	0.90	2.30	3.04	7.74	121	120	68	62	21

Matrix effect (%)=(100×peak area in spiked urine/peak area in spiked water)−100

Matrix effects >0 implies ion enhancement

Matrix effects <0 implies ion suppression

Table 4 Concentrations (μg kg^{-1}) of perfluorinated acids in human hair

	PFBA	PFOA	PFNA	PFDA	PFBS	PFHxS	PFOS	FDEA
1	<MLOQ	0.1	<MLOQ	<MLOD	<MLOD	<MLOD	<MLOQ	<MLOQ
2	39.3	<MLOD	<MLOD	<MLOQ	<MLOD	<MLOD	<MLOQ	<MLOD
3	<MLOQ	<MLOQ	<MLOD	<MLOQ	<MLOD	<MLOD	<MLOD	<MLOD
4	5.9	<MLOD	1.3	<MLOQ	<MLOD	<MLOD	5.8	<MLOD
5	<MLOQ	<MLOQ	<MLOD	<MLOQ	<MLOD	<MLOD	<MLOD	<MLOD
6	<MLOQ	<MLOQ	<MLOD	<MLOD	<MLOQ	<MLOD	<MLOQ	<MLOD
7	<MLOQ	0.4	<MLOD	<MLOD	<MLOD	<MLOQ	<MLOQ	<MLOD
8	<MLOQ	<MLOQ	<MLOD	<MLOQ	<MLOQ	<MLOQ	4.3	<MLOD
9	<MLOQ	<MLOQ	<MLOD	<MLOQ	<MLOD	<MLOQ	<MLOQ	<MLOQ
10	<MLOD	2.6	<MLOD	<MLOQ	<MLOQ	<MLOD	3.7	<MLOD
11	<MLOQ	<MLOQ	<MLOD	<MLOQ	<MLOD	<MLOQ	6.2	<MLOQ
12	38.1	1	<MLOQ	<MLOD	<MLOD	13.3	<MLOD	<MLOQ
13	<MLOQ	<MLOQ	<MLOQ	<MLOD	<MLOD	<MLOD	<MLOD	<MLOD
14	<MLOQ	<MLOQ	<MLOQ	<MLOD	<MLOD	<MLOD	<MLOQ	<MLOD
15	<MLOQ	1.3	0.5	<MLOD	<MLOD	10.4	5.1	<MLOD
16	<MLOQ	<MLOD	<MLOD	<MLOQ	<MLOD	<MLOQ	<MLOQ	3.2
17	<MLOQ	0.5	<MLOD	<MLOD	<MLOD	<MLOQ	<MLOQ	<MLOQ
18	<MLOQ	<MLOQ	<MLOQ	<MLOQ	<MLOD	<MLOQ	7.2	<MLOD
19	<MLOQ	<MLOQ	<MLOD	<MLOD	10.7	5.6	<MLOD	<MLOD
20	<MLOQ	<MLOQ	<MLOQ	<MLOD	<MLOQ	<MLOD	4.8	5.6
21	<MLOD	<MLOQ	0.5	<MLOQ	0.8	10.7	4.2	4.3
22	8.2	0.7	<MLOD	45.9	<MLOD	<MLOD	6.5	<MLOD
23	28.3	6.1	<MLOD	<MLOQ	<MLOD	10.8	6.5	9.6
24	<MLOQ	<MLOQ	<MLOQ	<MLOQ	<MLOD	<MLOD	5.2	<MLOD

Levels of PFPeA, PFHxA, PFHpA, PFUdA, PFDoA, PFTrA, PFTeA, PFHxDA, PFODA, PFDS, FOSA, FHEA, FOEA were below the MLOD or even below the MLOQ

Urine samples were fortified with 10 μL surrogate internal standard solution to furnish a final concentration of 20 μg L^{-1} of each standard. Precipitation of traces of proteins (150 μL of each sample) was induced by mixing the samples with acetonitrile (1:1). After centrifugation at 4000 rpm for 10 min, 100 μL of the supernatant was transferred to a PP insert vial. FHEA, FOEA and FDEA were not analyzed in urine samples because of lack reproducibility for these compounds in urine.

Chromatographic conditions and detection

The gradient used is given in Table 1. The solvent system used was:

- loading pump A: 0.1% formic acid in water; B: acetone–isopropanol–acetonitrile 10:45:45; C: water; D: methanol
- eluting pump E: NH_4Ac 20 mmol L^{-1} in water F: NH_4Ac 20 mmol L^{-1} in methanol

Acquisition was performed in selected reaction monitoring mode (SRM). Precursor ions, transitions used for quantification and confirmation, and retention times of the target compounds are listed in Table 2. The total run time for each injection was 16 min and the injected volume was 20 μL.

For identification purposes, retention times of PFCs in the standards and in the samples were compared at a tolerance of ±2.5%. Moreover, in accordance with the 2002/657/EC Decision, the relative ion intensities (ratios between the areas of the most intense transitions) of a compound in a sample or in a matrix were compared with the same ratios in the calibration curve, with a tolerance of ±2 SD.

Quantification was performed by use of surrogate internal standards. The ratio of the most intense transition peak area to the corresponding surrogate peak area was plotted against the concentration.

Method validation

The method was validated in accordance with the criteria described in EU Commission Decision 2002/657/EC [19], with special attention to linearity, recovery, matrix effects, precision, limits of detection (LOD) and quantification (LOQ), decision limit (CCα), and detection capability (CCβ). Decision limit (CCα) was established by analysis of 20 blanks and detection capability (CCβ) as $CC\beta = CC\alpha + 1.64 \times SD_{CC\alpha}$. Linearity was based on

Table 5 Levels of PFCs in urine samples higher than the MLOQ

	Concentration (ng L^{-1}) (RSD, %, n=3)							
Sample	PFBA	PFHxA	PFHpA	PFOA	PFDA	PFUdA	PFBS	PFHxS
Urine 1	100907 (13)	<MLOQ	<MLOD	<MLOQ	<MLOD	<MLOD	<MLOQ	3980 (20)
Urine 2	52716 (17)	<MLOQ	<MLOQ	<MLOD	<MLOD	<MLOD	<MLOQ	<MLOQ
Urine 3	464296 (20)	1105 (30)	<MLOD	1796 (17)	<MLOD	<MLOD	<MLOQ	<MLOQ
Urine 4	1495676 (16)	49675 (10)	4612 (20)	132 (10)	<MLOD	<MLOQ	<MLOQ	14753 (7)
Urine 5	63468 (15)	<MLOQ	4969 (13)	<MLOQ	<MLOD	<MLOD	<MLOQ	39270 (10)
Urine 6	1103185 (3)	<MLOQ	<MLOQ	348 (4)	<MLOD	<MLOD	<MLOQ	329 (25)
Urine 7	752665 (6)	<MLOQ	<MLOQ	1362 (19)	<MLOD	<MLOD	<MLOQ	<MLOQ
Urine 8	155845 (7)	<MLOD	<MLOQ	1353 (28)	<MLOD	<MLOD	<MLOQ	<MLOQ
Urine 9	120802 (19)	<MLOQ	<MLOD	830 (25)	<MLOD	<MLOD	<MLOQ	<MLOQ
Urine 10	133268 (25)	<MLOQ	<MLOQ	124 (6)	<MLOD	<MLOD	<MLOQ	<MLOQ
Urine 11	769376 (28)	<MLOQ	861 (29)	875 (13)	<MLOD	<MLOD	<MLOQ	2499 (23)
Urine 12	334750 (2)	<MLOD	592 (21)	<MLOQ	<MLOD	<MLOD	<MLOQ	9167 (4)
Urine 13	1131864 (5)	<MLOD	<MLOD	<MLOQ	670 (29)	5073 (18)	<MLOQ	43081 (12)
Urine 14	937200 (6)	<MLOQ	969 (19)	770 (21)	<MLOD	<MLOQ	<MLOQ	82744 (5)
Urine 15	797286 (2)	<MLOQ	<MLOQ	1140 (5)	919 (14)	2043 (10)	<MLOQ	<MLOQ
Urine 16	129905 (8)	<MLOD	<MLOD	<MLOQ	<MLOD	<MLOQ	<MLOQ	533 (24)
Urine 17	162323 (10)	<MLOD	<MLOQ	309 (11)	<MLOD	<MLOQ	<MLOQ	<MLOQ
Urine 18	498718 (4)	<MLOQ	<MLOQ	214 (23)	<MLOD	<MLOD	<MLOQ	12628 (10)
Urine 19	1017116 (6)	<MLOQ	<MLOD	2890 (14)	<MLOD	<MLOQ	<MLOQ	429 (27)
Urine 20	1478447 (6)	<MLOQ	<MLOQ	664 (30)	<MLOD	<MLOD	<MLOQ	<MLOQ
Urine 21	177139 (5)	<MLOQ	<MLOQ	<MLOQ	<MLOD	<MLOD	<MLOQ	<MLOQ
Urine 22	169728 (21)	<MLOD	<MLOQ	<MLOQ	<MLOD	<MLOD	<MLOQ	<MLOQ
Urine 23	580576 (20)	<MLOQ	<MLOD	496 (26)	<MLOD	<MLOQ	<MLOQ	<MLOQ
Urine 24	339182 (7)	<MLOD	<MLOD	<MLOQ	<MLOD	<MLOQ	<MLOQ	693 (20)
Urine 25	153123 (8)	<MLOQ	<MLOQ	<MLOQ	658 (9)	<MLOQ	<MLOQ	<MLOQ
Urine 26	131184 (9)	<MLOD	<MLOD	487 (26)	721 (17)	<MLOD	<MLOQ	<MLOQ
Urine 27	353720 (17)	<MLOQ	<MLOQ	<MLOD	<MLOD	<MLOD	<MLOQ	21832 (9)
Urine 28	700399 (17)	39421 (25)	3394 (23)	214 (18)	<MLOD	<MLOD	<MLOQ	17282 (3)
Urine 29	123444 (8)	<MLOD	<MLOQ	<MLOQ	<MLOD	<MLOQ	<MLOQ	2469 (17)
Urine 30	78014 (9)	<MLOQ	<MLOQ	<MLOQ	<MLOD	<MLOD	1320 (33)	<MLOQ
average	483545	30068	2567	824	743	3558	1050	11889
SD	435363	25600	2000	722	121	2143	58	20027
Median	336966	39422	2182	664	696	3558	1041	2484

Levels of PFPeA, PFNA, PFDoA, PFTrA, PFTeA, PFHxDA, PFODA, PFOS, PFDS and PFOSA were below the MLOD or even below the MLOQ

seven point matrix-matched calibration curves ranging from 0.05 to 500 ng g^{-1} for hair and from 0.05 to 500 ng mL^{-1} for urine. A chromatogram obtained from blank hair fortified with 10 μg kg^{-1} PFCs in a standard mixture is presented in Fig. 1. Repeatability (intra-day precision) and reproducibility (inter-day precision) were calculated as the average percentage relative standard deviation (RSD, %) for fortified blank samples of urine and hair (twenty replicates, 150 μL urine and 0.25 g hair per replicate) at concentrations equivalent to 5 ng mL^{-1}. For intra-day precision the different replicates of the fortified samples were analysed on three different occasions on the same day; for inter-day precision the extracts were prepared on three consecutive days.

Results and discussion

Methods development

To minimize as much as possible the handling of the samples before the turbulent flow chromatography, while at the same time eliminating any major matrix effects, the hair sample pre-treatment procedure was optimized and four different approaches were evaluated:

1. alkaline digestion [9] using methanol containing 10 mmol L^{-1} NaOH;
2. acid digestion [20] with water at pH 3.4;

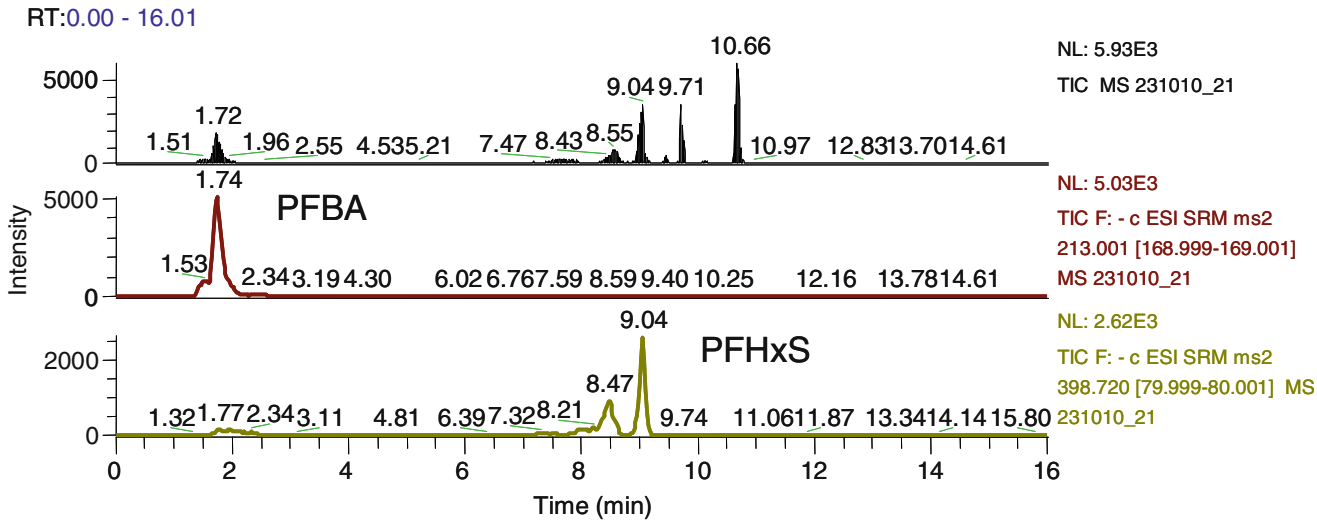

Fig. 2 TIC (total ion chromatogram) obtained from analysis of urine sample no. 4. In this sample two analytes were found: PFBA and PFHxS

3. neutral extraction with acetonitrile in an orbital digester at 125 u min^{-1}; and
4. ultrasonic extraction with acetonitrile.

In each case, different solvent volumes were tested (2, 2.5, 4, 4.5, and 5 mL). For most of the compounds the best recoveries were obtained by use of 5 mL acetonitrile and extraction and ultrasonication for 15 min.

Four steps were considered when developing the TFC method. The first was a loading step and involved loading of the sample and sufficient washing of the TFC column to remove the matrix. Water acidified with formic acid at pH 3.4 and subsequent washing with water was proved to be the optimum sequence for retention of most of the analytes and elimination of the matrix interferences. The second step was a transfer step in which the analytes of interest were desorbed from the TFC columns on to the analytical column. A 70:30 mixture of water acidified with formic acid at pH 3.4 and methanol was proved to be the most suitable mobile phase composition to elute the analytes from the TFC columns. The third step was chromatographic separation on the analytical LC column by use of an elution gradient. The last step was equilibration in which the initial conditions were set for the next run. To avoid carry-over effects, different solvent compositions were optimized for the TFC columns wash steps.

Identification and quantification

Instrumental mass spectral conditions for some of the compounds included in this study were those optimized in previous work by our group [21], whereas for fluorotelomer acids the mass spectral instrumental conditions were optimized by direct infusion of the standards using the negative ionization mode. Collision-induced dissociation (CID) mass spectra were recorded for each analyte and the optimum collision energies were determined for the selected ion transitions. Two transitions were monitored for each analyte, as indicated in Table 2.

Method validation

The method was validated in accordance with the EU requirements stated in the commission decision 2002/657/EC [22].

For quantification purposes surrogate internal standards were always added at concentrations of 20 and 5 μg L^{-1} for hair and urine, respectively, before sample pre-treatment.

The results shown in Table 3 indicate inter day repeatability ranged from 5–19% for urine and 8–26% for hair, and was below 20% for most of the compounds in both matrices. It is, maybe, worth noting that the compounds with the lowest and highest values for urine were not the same as those for hair.

Limits of detection (LODs) for the 21 PFCs, were defined as three and nine times the signal-to-noise (*S*/*N*) value for the quantifier ion of each analyte from analysis of six aliquots of blank urine and hair fortified at 5 ng g^{-1} or 5 ng mL^{-1} for hair and urine, respectively, and confirmed by dilution for each analyte. The MLOD for each analyte in each type of sample are listed in Table 3. It is apparent MLOD were in the ranges 0.01 to 2.7 ng mL^{-1} and 0.01 to 4 ng g^{-1} for urine and hair, respectively. MLOQ were from 0.1 to 9 ng mL^{-1} for urine and from 0.04 to 13.4 ng g^{-1} for hair. It is apparent MLOD were higher for longer-chain PFCs than for shorter-chain. This was in agreement with the mass of every compound because this difference between the shorter and the longer PFC (at least 600 Da) implies, for the same concentration, fewer molecules of PFC.

Decision limit (CCα) and detection capability (CCβ) values were calculated according to the EU regulation.

CCα was evaluated by analyzing 20 blank samples (not fortified) and calculating the signal-to-noise ratio for the time window in which the analyte is expected. Three times the signal-to-noise ratio was used as the decision limit with 5% certainty. CCβ values were obtained from analysis of approximately 20 blank samples, this time fortified at the CCα concentrations, where possible, or close to them. The corresponding concentration at the CCα value plus 1.64 times the standard deviation of the within-laboratory reproducibility of the mean measured content at the decision limit equals the detection capability (β=5 %). These limits are listed in Table 3 for each compound and matrix.

Calibration series were prepared in LC water, hair, and urine to calculate recovery and to address matrix ionization suppression or enhancement effects. To ensure homogeneous distribution of the analytes in the matrix, after fortification, the samples were mixed on a rotary mixer for 10 min then left to equilibrate for 1 h at room temperature. Strong ion suppression and enhancement effects were observed for the extracts of urine and hair, as can be seen from the percentage matrix effect for urine spiked at 10 ng mL^{-1} compared with water spiked at the same concentration (Table 3). Ion suppression and enhancement for urine was higher than was observed for hair.

Average recoveries of PFCs at different concentrations are shown in Table 3. For hair samples, the results showed that for most of the compounds recoveries were in the range 60–90%; for urine recoveries were in the range 65–105%.

Analysis of human samples

Concentrations higher than the MLOQ for selected PFCs in human hair samples are listed in Table 4. It is apparent 75% of the samples were positive for at least one of the target compounds although no compound was found in 100% of the hair samples. PFOS and PFOA were the compounds more frequently found. PFOA was present at concentrations in the range 0.1–6 ng g^{-1} among the positive samples, whereas PFOS was between 3.7 and 7 ng g^{-1} for samples in which concentrations were higher than the MLOQ. It should be also pointed out that some of the less frequently found compounds were present at high concentrations. For example, PFBS was present in five samples but at concentrations between 6 and 39 ng g^{-1}; this compound was found at the highest level. Other compounds, for example PFHxS and PFNA were also present in some samples. Because of the lack of data for PFCs in human hair these results cannot be compared with those from similar studies. However, other studies report the accumulation in other human matrices, for example liver. In these matrices the same compound profile was found. For example, Kärrman et al., [23] studied the presence of PFCs in human liver samples and found the same pattern of PFCs but at much higher concentrations than in our study. In a large study carried out by the group of Calafat et al. [14] concentrations of PFCs were reported for serum samples; the results from this study also had a similar profile.

In urine samples 44% of the 18 analysed PFCs were present at positive levels. PFBA was found in 100% of the samples at the highest levels. The other PFCs higher than the MLOQ were PFHxA, PFHpA, PFDA, PFUnA, and PFBS, at concentrations from below the reporting limit to 49,675 ng L^{-1}, even though most of the samples were below the MLOQ. Concentrations of PFCs higher than the MLOD and MLOQ are listed in Table 5. An example of a chromatogram from a urine sample is given in Fig. 2. Completed results of urine samples are reported in Electronic Supplementary Material Table S2.

Concentrations of PFOA and PFHxS ranged from below the reporting limit to 2,890 ng L^{-1} and 39,270 ng L^{-1}, respectively. PFBA values were from 52,716 to 1,495,676 ng L^{-1}. During the analysis of this last compound, some of the samples were diluted because of the high concentrations in these samples.

The highest levels of PFBA could be directly related to elimination through urine. Another possible source could be biodegradation of other longer-carbon-chain compounds with PFBA as the end-product.

PFOA in urine is in agreement with previous studies [24]. Hundley et al. [24] found excretion was the major process of elimination of PFOA (120 h, approximately, after dose exposure) from rats and rabbits after accumulation by the blood and liver, followed by the kidneys, lungs, and skin to a lesser extent.

One of the most recalcitrant compounds in human blood is PFOS; it should, however, be remarked that this compound was not found in urine samples, in agreement with work performed by Olsen et al. [2] in which the elimination of this compound was, in most of the cases, longer than 4 years.

Regarding the presence of PFHxS, although this compound was at higher levels than the other compounds selected for this study the concentrations were lower than the concentrations previously reported for blood [2, 12, 25], suggesting that this compound has higher affinity for blood proteins.

Conclusions

A new method for automated multi-residue analysis of PFCs in urine and human hair is proposed. It combines turbulent flow chromatography, for extraction and purification of the analytes, with LC–MS–MS, for analytical separation and detection of the analytes. The advantages of this method are reduced sample preparation time and higher sample throughput. In addition, rapid and simple sample pre-treatment was optimized as a step before TFC. This simple pre-treatment consisted in solvent extraction and centrifugation, ensuring minimization of matrix effects.

MLOQ obtained were suitable for assessing the presence of the selected 21 PFCs in urine and hair.

These optimized approaches were applied to the analysis of 24 hair human hair samples and 30 urine samples. PFOS and PFOA were the compounds most frequently found in the hair samples analysed; the compounds found in urine were PFBA, PFOA, and PFHxS.

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